Department of Computer Science & Engineering

B.Tech · Third Year · Even Semester · 2025–26

Case Study Assignment

Text Mining, Spatial Database Mining & Web Mining

Apply Classification or Clustering to a Real-World Dataset

Course Code	Assigned On	Submission Date	Total Marks
CS-356	April 20, 2026	May 3, 2026	50 Marks

Student Details (Fill before submission)

Full Name: _____ **Roll Number:** _____

Section / Group: _____ **Domain Chosen:** _____

Domain options: Text Mining | Spatial Database Mining | Web Mining

1. Objective

This is an **individual** assignment. You are required to conduct a structured case study on **one** of the following data mining domains:

- **Text Mining** — discovering knowledge from unstructured textual data.
- **Mining Spatial Databases** — extracting patterns from geographic/location-based data.
- **Web Mining** — uncovering patterns from web content, structure, or usage data.

You must apply either a **Classification** or a **Clustering** algorithm to a real or publicly available dataset from your chosen domain, and present your findings as a written report with working code.

2. Choose Your Case Study Domain

Select **exactly one** domain from the three options below. Each option lists suggested problem statements to guide your topic selection.

Option A `Text Mining`

What it involves: Analysing raw text documents to extract patterns, topics, sentiment, or categories using NLP techniques.

Suggested Problem Statements:

- Sentiment analysis of Amazon product reviews (Classification).
- Topic modelling of BBC news articles using LDA (Clustering).
- Spam vs. ham email detection using Naïve Bayes (Classification).
- Fake news detection using TF-IDF + SVM (Classification).
- Clustering of research abstracts by subject area (Clustering).

Suggested datasets: IMDB Reviews, 20 Newsgroups, SMS Spam Collection, AG News (Kaggle / UCI ML Repository / NLTK corpora).

Option B `Mining Spatial Databases`

What it involves: Mining geographic, location-tagged, or coordinate-based data to find spatial patterns, clusters, or classes.

Suggested Problem Statements:

- Crime hotspot detection in a city using DBSCAN (Clustering).
- Land-use classification from satellite/GIS raster data (Classification).
- Clustering of hospital/school locations by accessibility (Clustering).
- Road traffic accident zone analysis and risk classification (Classification).
- Earthquake epicentre clustering by magnitude and depth (Clustering).

Suggested datasets: Chicago Crime Data, USGS Earthquake Catalog, NASA SEDAC, OpenStreetMap, India GIS Open Data.

Option C `Web Mining`

What it involves: Extracting useful information from web content (pages, hyperlinks) or user interaction logs (clickstreams, sessions).

Suggested Problem Statements:

- Web page topic classification using content features (Classification).
- User session clustering from clickstream data (Clustering).
- Social network community detection on Twitter follower graphs (Clustering).
- Malicious URL classification using lexical features (Classification).
- E-commerce product recommendation via user-behaviour clustering (Clustering).

Suggested datasets: MSWEB Dataset (UCI), KDD Cup 2000 Clickstream, PhishTank URLs, Stanford SNAP social graph datasets.

Important: Two students in the same section may not use the same dataset. Confirm your dataset with the instructor within **5 days** of receiving this assignment.

3. Tasks

Complete all six tasks in sequence. Each task must be documented in the corresponding section of your report.

Task 1 — Problem Definition `[05 M]`

Clearly state:

- The domain (Text / Spatial / Web Mining).
- The specific problem or question you are addressing.
- Why this problem is significant in the real world.
- Whether you will use Classification or Clustering, and a one-paragraph justification.

Task 2 — Dataset Collection & Description [06 M]

- Source URL and citation of the dataset.
- Number of instances and attributes.
- Attribute types (categorical, numerical, textual, spatial).
- Class distribution (for classification) or expected number of clusters.
- A sample of 5 rows presented as a table.

Task 3 — Preprocessing & Feature Engineering [07 M]

- Handle missing values, outliers, and duplicate records.
- Normalise or standardise numerical features where necessary.
- *Text Mining:* Tokenisation, stop-word removal, stemming / lemmatisation, TF-IDF or word embeddings.
- *Spatial:* Coordinate reprojection, extraction of distance/area features.
- *Web Mining:* URL parsing, session segmentation, graph feature extraction.
- Report the final feature vector dimensionality used as input to the algorithm.

Task 4 — Algorithm Implementation [12 M]

Choose and implement one algorithm from Section 4. Your implementation must:

- Be written in Python (scikit-learn, NLTK/spaCy, geopandas, NetworkX, etc.).
- Include inline code comments explaining each major step.
- Include hyperparameter choices and a brief justification (e.g., $k = 5$ in k-NN, $k = 4$ in K-Means).
- For classification: use an 80/20 train-test split.
- For clustering: apply the Elbow Method or Silhouette analysis to select optimal parameters.

Task 5 — Evaluation & Visualisation [10 M]

Report all applicable metrics from Section 4 and include **at least two** of the following plots:

- Confusion matrix heatmap (classification).
- ROC/AUC curve (binary classification).
- Cluster scatter plot with centroids (K-Means / GMM).
- Dendrogram (hierarchical clustering).

- Word cloud per cluster or class (text mining).
- Spatial map showing clusters or class boundaries (spatial mining).
- Elbow or Silhouette score plot (clustering).

Task 6 — Analysis & Conclusion [10 M]

- Interpret your results: what patterns or insights did you discover?
- Compare performance to a baseline (majority-class classifier or random clustering).
- Discuss at least two limitations of your dataset or methodology.
- Propose one concrete future improvement or extension.

4. Allowed Methods & Evaluation Metrics

Classification Algorithms (choose one)

Algorithm	Notes
Naïve Bayes	Particularly suitable for text mining (Multinomial NB).
Decision Tree	Good for interpretability; show the tree diagram.
Random Forest	Ensemble method; report feature importances.
Support Vector Machine	Effective for high-dimensional text feature spaces.
k-Nearest Neighbours	Useful for spatial data; vary k and report results.
Logistic Regression	Strong baseline; report coefficient magnitudes.

Metrics: Accuracy · Precision · Recall · F1-Score · Confusion Matrix · ROC-AUC

Clustering Algorithms (choose one)

Algorithm	Notes
K-Means	Apply Elbow Method to choose k; report inertia.
Hierarchical (Agglomerative)	Include dendrogram; try Ward, Complete, Average linkage.
DBSCAN	Ideal for spatial data; vary `eps` and `min_samples`.
Gaussian Mixture Model	Soft clustering; report log-likelihood and BIC.

Metrics: Silhouette Score · Davies–Bouldin Index · Calinski–Harabasz Index · Inertia (Elbow)

5. Deliverables & Submission

#	Item	Description	Marks
D1	**Written Report**	PDF, min. 8 pages. Structure: Title Page → Abstract → Introduction → Dataset → Preprocessing → Methodology → Results → Conclusion → References (IEEE format).	20
D2	**Source Code**	`.ipynb` or `.py` with inline comments. Include `requirements.txt`. Code must be reproducible on a fresh environment.	15
D3	**Visualisations**	Min. 2 plots embedded in the report and saved separately as `.png` files.	08
D4	**Viva / Presentation**	5-minute individual oral viva covering approach and results.	07
		Total	**50**

Submit a single ZIP archive named:

`RollNo_Name_DM_CaseStudy.zip`

Upload to the course portal before **May 3, 2026, 11:59 PM**.

6. Marking Scheme

Component	Marks
Problem definition, domain motivation, and algorithm justification	08
Dataset description and exploratory data analysis	05
Data preprocessing and feature engineering	08
Correct and well-commented algorithm implementation	12
Evaluation metrics reported correctly with visualisations	10
Quality of analysis, insights, and conclusion	07
Viva clarity and ability to answer questions	05
Total	**50**

7. Important Guidelines

1. **Individual work only.** Sharing of code, notebooks, or reports between students will result in **zero marks** for all parties and may attract disciplinary action.

2. **Dataset citation.** Every dataset must be properly cited in IEEE format in your references section. Provide the direct URL.

3. **Plagiarism check.** All reports will be submitted to a plagiarism detection tool. A

similarity score above **20%** will result in rejection and a request for resubmission with a penalty of 10 marks.

4. **Late submission penalty.** 5 marks will be deducted per day after the deadline. No submission will be accepted after **7 days** beyond the due date.

5. **Recommended tools.** Python 3.8+ with: `pandas`, `numpy`, `scikit-learn`, `matplotlib`, `seaborn`, `nltk` / `spacy` (text mining), `geopandas` / `folium` (spatial), `networkx` / `beautifulsoup4` (web mining).

6. **Report language.** The report must be written in clear, formal English. Bullet-point-only reports will not be accepted; use complete sentences and paragraphs.

7. **Queries.** Email queries to the course instructor at least **5 days** before the submission deadline. Last-minute queries will not be addressed.

Evaluation Rubric (Quick Reference)

Criterion	**Excellent (Full)**	**Satisfactory (60%)**	**Poor (<40%)**
Problem definition	Clear, specific, well-motivated	Somewhat vague	Missing or generic
Dataset & EDA	Rich EDA with plots	Basic statistics only	No EDA
Preprocessing	All steps applied and justified	Partial preprocessing	Raw data used directly
Implementation	Correct, commented, reproducible	Minor errors, runs	Incorrect or incomplete
Evaluation	All metrics + 2+ plots	One metric only	No evaluation
Analysis	Deep insights, honest limitations	Surface-level discussion	Copy of results only

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